Q1 · Predicting market betas

Historical beta forecasts contain information about subsequent exposure, but one-month realized betas remain noisy. Welch's slope-winsorized estimator has lower out-of-sample (OOS) error than the OLS benchmarks on both weighting schemes. The proposed shrinkage rule does not improve on Welch (Tables 1–3; Figures A–B).

Historical estimation and predictive calibration

Using the PS1 common-stock universe, I estimate intercept market models from daily stock excess returns and CRSP value-weighted total market excess returns, subtracting the same-date French risk-free rate. Forecasts use observations through actual month-end t; realized beta is intercept OLS over month t+1. PS1 month-start dates are observation-month labels, not information-availability dates. The historical OLS windows are one year, three years and expanding since 1960. The last is an explicitly approved approximation to the assignment's full-available-history benchmark; pre-1960 returns are unavailable here.

$$r^{e}_{i,d} = \alpha_i + \beta_i \, r^{e}_{m,d} + \varepsilon_{i,d}$$

Welch clips each excess stock return to the interval between −2x and 4x, where x is the contemporaneous market excess return, and estimates an intercept and slope with exponential age weights exp(−ρj). The Appendix value ρ=2/256 is primary; 2/252, stated in the paper's introduction, and valid-pair versus market-calendar aging are sensitivities (Table 3). The main specification ages market sessions. Histories require 20 valid daily pairs; Welch also requires a valid final-session return. Realized targets require at least max(10, half the month's market sessions rounded up).

$$\beta^{real}_{i,t+1} = \gamma_0 + \gamma_1 \, b_{i,t} + u_{i,t+1}$$

The primary calibration is pooled OLS on matched stock-months (Table 1). OOS Welch has intercept −0.0262, slope 1.0911 and R^2=0.1335; one-year OLS has slope 0.7267 and R^2=0.1110. Figure B contrasts saved decile averages with the pooled fitted line and a 45-degree reference. Noisy historical regressors can attenuate predictive slopes, while noisy future-beta targets reduce explanatory power. Changing business mix, leverage and market conditions can also make old exposures stale; calibration is not a claim that true beta is constant.

Forecast accuracy and the proposed improvement

The common sample has 3,397,913 stock-months over July 1963–December 2025. EW weights are equal within month; VW weights use positive month-end-t market capitalization. Monthly MSEs are averaged equally before taking their square root. Table 2 and Figure A show OOS VW RMSE of 0.5719 for Welch, versus 0.5841, 0.6047 and 0.6539 for one-year, three-year and expanding OLS; EW Welch RMSE is 1.3794 versus 1.4074 for one-year OLS. Larger EW errors are consistent with greater influence from volatile small stocks.

$$\mathrm{RMSE} = \left[\, T^{-1} \sum_t \sum_i \omega_{i,t} \left(\beta^{real}_{i,t+1} - b_{i,t}\right)^2 \, \right]^{1/2}$$

I divide outcomes at July 2000, retaining the pre-existing PS1 split rather than selecting a favorable break: 444 IS months and 306 OOS months. Shrinkage $b^* = 1 + k(b^{Welch} - 1)$, $0 \le k \le 1$, is motivated by measurement error and possible mean reversion in market exposure. Minimizing IS monthly VW loss selects k=1 (Table 3). Frozen OOS, the rule equals Welch and has no improvement. Its exact paired loss difference and variance are zero, so its t-statistic is undefined; the saved roundoff-scale differences are not statistical evidence. No replacement was tuned on OOS outcomes.

Q2 · Forecasting factor and stock returns

Improving forecasts of one factor premium need not improve stock-return forecasts. In this sample, an expanding VAR(1) modestly improves OOS HML forecasts but worsens market and SMB forecasts. The resulting stock predictions have higher OOS MSE than expanding factor means, with a positive Newey–West paired-loss statistic (Tables 4–5; Figure C).

Joint exposures and historical-average benchmarks

Joint three-factor OLS uses the preceding 36 calendar months of stock excess returns and PS1 traded Mkt−RF, HML and SMB, with an intercept, 24 complete months and full rank. Forecasts use slopes available at t, excluding alpha. Correlated size/value factors explain why joint market exposure can differ from Q1's single-factor beta.

$$r^{e}_{i,s} = \alpha_{i,t} + b'_{i,t} F_s + \varepsilon_{i,s} \text{ ; forecast } r^{e}_{i,t+1} = b'_{i,t} \lambda_{t+1|t}$$

The first benchmark assigns every stock the expanding mean CRSP VW excess return. The second multiplies each exposure vector by expanding factor means, using observations through t only. The two factor-model forecasts share betas, samples and weights, isolating the contribution of premium forecasts.

An alternative forecast of the three premia

$$F_{s+1} = a + A F_s + \eta_{s+1} \text{ ; } \lambda_{t+1|t} = a_t + A_t F_t$$

The expanding VAR(1) uses lagged factor returns to capture persistence, reversals and cross-factor adjustment, although their signal may be weak. Its specification is fixed before OOS; coefficients update only with completed outcomes through t. This is not a valuation-spread model. Requiring 60 completed lagged training pairs with factors starting in July 1963 delays common evaluation to August 1968.

Matched IS/OOS results and uncertainty

The matched comparison contains 2,774,383 stock-months over 689 months: 383 IS and 306 OOS. Table 4 reports factor MSE and pseudo-R^2 relative to each expanding mean. OOS VAR pseudo-R^2 is −1.073% for the market, −2.046% for SMB and +1.038% for HML. The HML improvement is modest and should not be interpreted as strong evidence of a reliable premium forecast.

$$\text{Pseudo-}R^2 = 1 - \text{MSE}_{\text{model}} / \text{MSE}_{\text{benchmark}}$$

Table 5 reports stock outcomes on identical stocks, exposure snapshots and origin-ME weights. OOS VW MSE is 0.00850759 for the market mean, 0.00851672 for factor means and 0.00858404 for VAR premia, in squared decimal monthly returns. Factor means themselves trail the market-mean benchmark slightly. VAR's pseudo-R^2 is −0.790% against factor means and −0.899% against the market mean. EW comparisons point in the same direction; the table also retains the IS results.

Figure C cumulates the saved OOS monthly VW loss difference, VAR minus factor means. Its positive endpoint reflects deterioration, not an investment return. The average difference is 0.0000673, with Bartlett Newey–West(12) SE 0.0000322 and t=2.089 (EW t=2.047). These are descriptive normal-reference comparisons without a nested-model forecast-test correction. Extra premium-model parameters can amplify estimation noise, and correlated premium errors interact with stock exposures. The evidence therefore does not support this alternative as an improvement in stock-return prediction; no OOS replacement model was selected.

Q3 · Characteristic returns and measurement

Industry-controlled monthly regressions

Does value predict returns conditional on other characteristics? The full-sample coefficient is positive, but the OOS estimate is smaller and less precise (Table 6; Figure D). Monthly regressions use month-end-t characteristics and t+1 excess returns:

$$r^{e}_{t+1} = X_t f_{t+1} + u_{t+1} \; ; \; \min_f \Sigma_i \sqrt{ME}_{i,t} \left(r^{e}_{i,t+1} - X_{i,t} f\right)^2$$

X includes all represented FF12 industry indicators, without a separate intercept, and standardized log ME, log BM, profitability, investment and the origin Welch market-beta forecast from Q1. Characteristics use equal-weight population standard deviations on the matched monthly sample. The objective weight is √ME, so the equivalent least-squares row multiplier is $ME^{1/4}$. Market capitalization is current PERMNO ME in USD millions. Historical CRSP SIC is joined at calendar month-end and classified with the official FF12 ranges, rather than French's annually assigned industry portfolios. Missing or invalid SIC is excluded, not imputed to Other.

Annual PS1 fundamentals use prior-calendar-year statements and June formation: June-end origins can use the new formation vintage, while earlier months use the preceding June. BM retains PS1's prior-December company-ME denominator. Exact statement-release vintages are unavailable, so this is the inherited June-formation convention, not fully point-in-time accounting. The regression sample contains 2,668,791 stock-months over all 750 assignment months, with at least 849 stocks per month.

Economic magnitude and the weaker OOS value estimate

A characteristic coefficient is a monthly return associated with one cross-sectional standard deviation of that characteristic, conditional on the other regressors. The full-sample value coefficient is 0.1527% per month, with NW(12) SE 0.0447% and t=3.416. It describes a positive conditional association for cheaper stocks, not a causal effect. Profitability is also positive and investment negative in the full sample; log size and predicted beta have weak average coefficients (Table 6).

The value result is notably weaker OOS: the coefficient is 0.0781% per month, with SE 0.0700% and t=1.116, compared with an IS coefficient of 0.2040% and t=3.625. The later period alone does not give precise evidence of a positive conditional value premium. These estimates do not constitute a formal test of a structural break. Figure D shows monthly coefficients and a trailing 36-month mean, including the current month; it does not use a centered, forward-looking smoother.

Changing units versus changing the economic comparison

Re-expressing the five standardized characteristics as z_new=2z+3 preserves their information and fitted span. Slopes halve, while industry coefficients absorb translations because the full dummy span contains the constant. Table 7 records a maximum slope-mapping error of 2.984×10^{-15} and fitted-return difference of 2.225×10^{-13}; residual differences have the same magnitude since y is unchanged. These are numerical identities, not economic findings.

Omitting profitability instead changes the conditioning set. The saved no-profitability value coefficient averages 0.1453%, versus 0.1527% with full controls, on the same stocks, units and weights (Table 6). Other no-profitability coefficients are marked “Not estimated”: they were not retained in the accepted saved outputs, and no new regressions were run to fill the table. This limited comparison does not substitute full-model slopes for missing omitted-model estimates.

Q3 · Pure value and the source of the premium

Joint weighted residualization

I select standardized log BM as the value signal z. To isolate relative cheapness, C contains log ME, profitability, investment, predicted beta and all represented FF12 dummies. Removing these linear exposures separates value from the size, business characteristics and industry composition that often accompany it. The controls are projected jointly, using exactly the return-regression sample and WLS objective weights:

$$a = (C'WC)^{-1}C'Wz \; ; \; z^{\perp} = z - Ca \; ; \; C'Wz^{\perp} = 0$$

W=diag(√ME). The industry span contains the constant; an alternative control set without the full industry span would require an intercept. The residual is retained in its original projection units and is not standardized again. A positive residual identifies a stock with higher log BM than the selected controls predict, rather than necessarily a high absolute BM.

The controls explain an average 31.691% of weighted cross-sectional signal variation, and 30.115% OOS (Table 7). The maximum norm-scaled weighted-orthogonality error is 8.952×10^{-15}. This establishes negligible weighted linear exposure to the included controls, not independence from them or neutrality to omitted and nonlinear risks.

The FWL comparison: an identity, not evidence of a premium

$$r^{e}_{t+1} = C\delta + \gamma z + u \; ; \; r^{e}_{t+1} = \alpha^{\perp}1 + \gamma^{\perp}z^{\perp} + v$$

The weighted Frisch–Waugh–Lovell theorem implies $\gamma=\gamma^{\perp}$ when the projection and both return regressions share observations and weights, C spans the constant, and residual signal variation is nonzero. Orthogonality removes the control component from the signal's normal equation; it does not create a different conditional return slope. The largest observed coefficient difference is 4.267×10^{-16} (Table 7). Fitted returns and residuals from the two return regressions need not coincide because the second omits C.

Original value can bundle industry, size, profitability, investment and market-exposure effects. Pure value instead compares stocks whose chosen controls are similar: a positive pure signal means relatively greater cheapness after those linear comparisons. The unchanged conditional slope is true by construction; whether that slope is economically meaningful or persistent must be judged from the return estimates. In particular, the imprecise OOS value coefficient in Table 6 prevents the algebraic equality from being read as evidence of a robust later-period premium.

Scope of the conclusions

The assignment sample is July 1963–December 2025, subject to initialization and missing inputs. Q1 and Q3 cover all 750 outcome months; Q2 begins in August 1968 after common premium-model initialization. Expanding beta starts in 1960 rather than the full CRSP history, and accounting release-date vintages are unavailable. Welch decay and missing-session aging conventions are explicitly distinguished in Table 3. The attempted Q1 and Q2 alternatives do not improve OOS performance. These limitations qualify the findings; numerical WLS identities do not remove them.

References: Welch (2022), “Simply Better Market Betas,” Critical Finance Review 11(1), 37–64, especially Appendix A; Goyal and Welch (2008), “A Comprehensive Look at the Empirical Performance of Equity Premium Prediction,” Review of Financial Studies 21(4), 1455–1508; official Ken French Data Library daily RF and FF12 SIC definitions. Specification: Quantitative Investing Problem Set 2, pp. 1–5. Code and saved-source provenance accompany the submission in the separate reproducibility document.

Table 1. Q1 pooled predictive calibration

Period	Estimator	N	Intercept	Slope	R-squared
ALL	OLS: one year	3,397,913	0.2505	0.6459	0.0730
ALL	OLS: three years	3,397,913	0.1837	0.7299	0.0711
ALL	OLS: expanding since 1960	3,397,913	0.2115	0.7075	0.0556
ALL	Welch: 2/256, calendar	3,397,913	-0.0424	1.0226	0.0906
ALL	Welch: 2/252, calendar	3,397,913	-0.0413	1.0212	0.0906
ALL	Welch: 2/256, valid pairs	3,397,913	-0.0432	1.0236	0.0906
ALL	Welch: 2/252, valid pairs	3,397,913	-0.0421	1.0222	0.0906
ALL	Proposed shrinkage (k=1)	3,397,913	-0.0424	1.0226	0.0906
IS	OLS: one year	2,106,038	0.2435	0.5526	0.0466
IS	OLS: three years	2,106,038	0.1625	0.6501	0.0494
IS	OLS: expanding since 1960	2,106,038	0.1498	0.6492	0.0434
IS	Welch: 2/256, calendar	2,106,038	-0.0183	0.9182	0.0590
IS	Welch: 2/252, calendar	2,106,038	-0.0170	0.9165	0.0589
IS	Welch: 2/256, valid pairs	2,106,038	-0.0193	0.9196	0.0590
IS	Welch: 2/252, valid pairs	2,106,038	-0.0180	0.9179	0.0590
IS	Proposed shrinkage (k=1)	2,106,038	-0.0183	0.9182	0.0590
OOS	OLS: one year	1,291,875	0.2894	0.7267	0.1110
OOS	OLS: three years	1,291,875	0.2667	0.7776	0.0983
OOS	OLS: expanding since 1960	1,291,875	0.3610	0.7320	0.0694
OOS	Welch: 2/256, calendar	1,291,875	-0.0262	1.0911	0.1335
OOS	Welch: 2/252, calendar	1,291,875	-0.0257	1.0902	0.1336
OOS	Welch: 2/256, valid pairs	1,291,875	-0.0265	1.0916	0.1334
OOS	Welch: 2/252, valid pairs	1,291,875	-0.0260	1.0907	0.1335
OOS	Proposed shrinkage (k=1)	1,291,875	-0.0262	1.0911	0.1335

Notes: Unweighted pooled OLS with intercept: next-month realized beta on origin historical beta; all slopes and betas are unitless. ALL: July 1963-December 2025, 750 months; IS: July 1963-June 2000, 444; OOS: July 2000-December 2025, 306. Within each period every estimator uses exactly the same stock-months. Welch calendar and valid-pair aging are separate variants; expanding starts in 1960, not 1926. Shrinkage k=1 reproduces Welch. N counts stock-month observations, not months. Source: q1/pooled_calibration.csv.

Table 2. Q1 forecast accuracy

Estimator	IS EW	IS VW	OOS EW	OOS VW
OLS: one year	1.652447	0.738989	1.407397	0.584072
OLS: three years	1.639315	0.745945	1.411524	0.604677
OLS: expanding since 1960	1.644168	0.765454	1.438941	0.653876
Welch: 2/256, calendar	1.617226	0.727264	1.379353	0.571855
Proposed shrinkage (k=1)	1.617226	0.727264	1.379353	0.571855

Notes: RMSE in beta units equals sqrt(mean across months of normalized monthly MSE). EW normalizes equal stock weights within month; VW uses positive origin month-end ME. Identical common stocks across all seven baseline variants and the proposed rule, and identical stocks for EW/VW. IS N=2,106,038 in 444 months (July 1963-June 2000); OOS N=1,291,875 in 306 months (July 2000-December 2025). Market and stock daily excess returns subtract same-day French RF. Realized targets are next-month intercept OLS. The expanding estimate is explicitly since 1960. Source: q1/rmse.csv and monthly_losses_EW/VW.csv.

Interpretation

Welch outperforms the three OLS windows on OOS EW and VW RMSE. Large EW errors reflect greater influence of volatile small stocks and noisy one-month realized beta targets. The proposed IS-tuned shrinkage equals Welch because its constrained optimum is k=1; it does not improve OOS accuracy.

Table 3. Welch robustness and IS tuning

A. Decay and aging sensitivity: matched RMSE (beta units)

Welch specification	IS EW	IS VW	OOS EW	OOS VW
Welch: 2/256, calendar	1.617226	0.727264	1.379353	0.571855
Welch: 2/252, calendar	1.617273	0.727248	1.379222	0.571644
Welch: 2/256, valid pairs	1.617177	0.727264	1.379462	0.571861
Welch: 2/252, valid pairs	1.617223	0.727249	1.379329	0.571650

B. Shrinkage rule: 1 + k (Welch - 1); selected k = 1

IS trial k	IS VW MSE
0.00	0.6849124
0.25	0.6176455
0.50	0.5692232
0.75	0.5396457
1.00	0.5289127

C. Paired monthly loss: proposed minus Welch

Period / weights	Months	Saved mean loss	Saved NW SE	Meaningful t
IS / EW	444	8.002e-18	5.491e-18	undefined
OOS / EW	306	4.989e-18	3.423e-18	undefined
IS / VW	444	-1.375e-18	7.123e-19	undefined
OOS / VW	306	1.134e-19	4.023e-19	undefined

Notes: Appendix Welch rho=2/256 is primary; introduction rho=2/252 is a sensitivity. Objective age weights are exp(-rho*j). Calendar aging counts market sessions; pair aging counts usable pairs. Panel A shares Table 2 samples and dates. Panel B tunes only IS monthly VW MSE. Panel C preserves the saved roundoff-scale loss differences and Bartlett NW(12) SEs; with k=1 the forecasts are algebraically identical. The true difference/variance are zero, so t is undefined, not zero. Finite ratios in the original saved CSV are floating-point artifacts and are not statistical evidence. Further minimum-history sensitivities are retained in the source package. Sources: q1/rmse.csv, IS_shrinkage_trials.csv, paired_losses_NW.csv.

Table 4. Q2 factor-premium forecasts

Period	Factor	Forecast	Months	MSE	Pseudo-R2 (%)
IS	Mkt-RF	Expanding mean	383	0.00212692	0.000
IS	Mkt-RF	VAR(1)	383	0.00215449	-1.296
IS	SMB	Expanding mean	383	0.00107326	0.000
IS	SMB	VAR(1)	383	0.00105873	1.354
IS	HML	Expanding mean	383	0.00081509	0.000
IS	HML	VAR(1)	383	0.00081036	0.581
OOS	Mkt-RF	Expanding mean	306	0.00204910	0.000
OOS	Mkt-RF	VAR(1)	306	0.00207109	-1.073
OOS	SMB	Expanding mean	306	0.00076127	0.000
OOS	SMB	VAR(1)	306	0.00077684	-2.046
OOS	HML	Expanding mean	306	0.00112341	0.000
OOS	HML	VAR(1)	306	0.00111174	1.038

Notes: MSE is in squared decimal monthly returns; pseudo-R2 = 100*(1-MSE_model/MSE_expanding_mean) in percent. Expanding mean is the benchmark for each factor, so its pseudo-R2 is zero by definition. IS: August 1968-June 2000, 383 matched factor-months; OOS: July 2000-December 2025, 306. Each premium uses identical outcome months across mean and VAR(1); there is one realization per factor per month. PS1 traded factors start July 1963, and the 60 completed lagged-pair minimum delays common evaluation to August 1968. All premium fits and averages use outcomes available through origin t only. Source: q2/factor_metrics.csv, verified against premium_forecasts.csv.

Interpretation

The OOS VAR improves HML MSE modestly but worsens market and SMB MSE. Better accuracy in one premium does not imply better individual-stock forecasts because exposure vectors and correlated premium forecast errors determine stock forecast losses.

Table 5. Q2 stock-return forecast accuracy

Period / weight	Forecast	MSE	R2: market (%)	R2: factors (%)
IS / EW	Historical market mean	0.02912968	0.000	0.461
IS / EW	Expanding factor means	0.02926466	-0.463	0.000
IS / EW	Expanding VAR(1)	0.02932776	-0.680	-0.216
IS / VW	Historical market mean	0.00826509	0.000	0.489
IS / VW	Expanding factor means	0.00830574	-0.492	0.000
IS / VW	Expanding VAR(1)	0.00839646	-1.589	-1.092
OOS / EW	Historical market mean	0.04002397	0.000	0.091
OOS / EW	Expanding factor means	0.04006050	-0.091	0.000
OOS / EW	Expanding VAR(1)	0.04022985	-0.514	-0.423
OOS / VW	Historical market mean	0.00850759	0.000	0.107
OOS / VW	Expanding factor means	0.00851672	-0.107	0.000
OOS / VW	Expanding VAR(1)	0.00858404	-0.899	-0.790

Notes: MSE uses squared decimal monthly excess returns, averaging normalized monthly losses equally. Pseudo-R2 is shown in percent against two explicit benchmarks: historical CRSP VW excess-return mean, or expanding factor means. IS N=1,618,939 stock-months in 383 months (August 1968-June 2000); OOS N=1,155,444 in 306 months (July 2000-December 2025). All three forecasts share stocks, outcomes, origin ME and factor exposure snapshots; EW and VW use the same keys. Joint 36-month three-factor OLS requires 24 valid months. Alpha is excluded. Sources: q2/stock_metrics.csv and monthly_stock_losses_EW/VW.csv.

OOS paired uncertainty

Weights / comparison	Mean loss difference	NW(12) SE	t-statistic
EW VAR - factor means	0.0001693	0.0000827	2.047
VW VAR - factor means	0.0000673	0.0000322	2.089

Positive paired differences mean worse performance for VAR. NW statistics are descriptive normal-reference comparisons without a nested-model forecast-test correction.

Table 6. Q3 mean characteristic returns

Period	Specification / characteristic	Mean (%)	NW SE (%)	t-statistic
ALL	Full: Log ME	-0.0495	0.0715	-0.693
ALL	Full: Log BM	0.1527	0.0447	3.416
ALL	Full: Profitability	0.0715	0.0212	3.372
ALL	Full: Investment	-0.0867	0.0158	-5.492
ALL	Full: Predicted market beta	0.0155	0.0748	0.208
ALL	Without OP: Log BM	0.1453	0.0439	3.312
IS	Full: Log ME	-0.0985	0.1036	-0.951
IS	Full: Log BM	0.2040	0.0563	3.625
IS	Full: Profitability	0.0780	0.0293	2.666
IS	Full: Investment	-0.1103	0.0186	-5.928
IS	Full: Predicted market beta	0.0461	0.0805	0.573
IS	Without OP: Log BM	0.1956	0.0555	3.527
OOS	Full: Log ME	0.0215	0.0921	0.234
OOS	Full: Log BM	0.0781	0.0700	1.116
OOS	Full: Profitability	0.0620	0.0300	2.068
OOS	Full: Investment	-0.0524	0.0264	-1.986
OOS	Full: Predicted market beta	-0.0289	0.1443	-0.200
OOS	Without OP: Log BM	0.0723	0.0684	1.058

Unavailable no-profitability coefficients: ALL, IS and OOS

Without-OP characteristic	Mean (%)	NW SE (%)	t-statistic
Log ME	Not estimated	Not estimated	Not estimated
Investment	Not estimated	Not estimated	Not estimated
Predicted market beta	Not estimated	Not estimated	Not estimated

Notes: Means and SEs are percent monthly excess return per cross-sectional population SD (both x100). Full model: five characteristics plus full FF12 dummies, no separate intercept. Objective weights sqrt(ME); row scaling ME^0.25. ALL: July 1963-December 2025, 750 months, N=2,668,791 stock-months; IS: 444 months; OOS: 306. Equal-time means and Bartlett NW(12) SEs, no finite-sample correction. No-OP value uses identical stocks, units and weights; only its saved coefficient series is aggregated. Sources: q3/mean_coefficients_NW.csv and omitted_profitability.csv.

"Not estimated" applies to ALL/IS/OOS in this revision: those coefficients were not retained in the accepted saved outputs. No new regressions are run and no full-model slopes are substituted. Profitability is excluded by definition. Accounting uses prior-calendar-year statements and June formation, without historical release-date vintages.

Table 7. Pure-value and numerical diagnostics

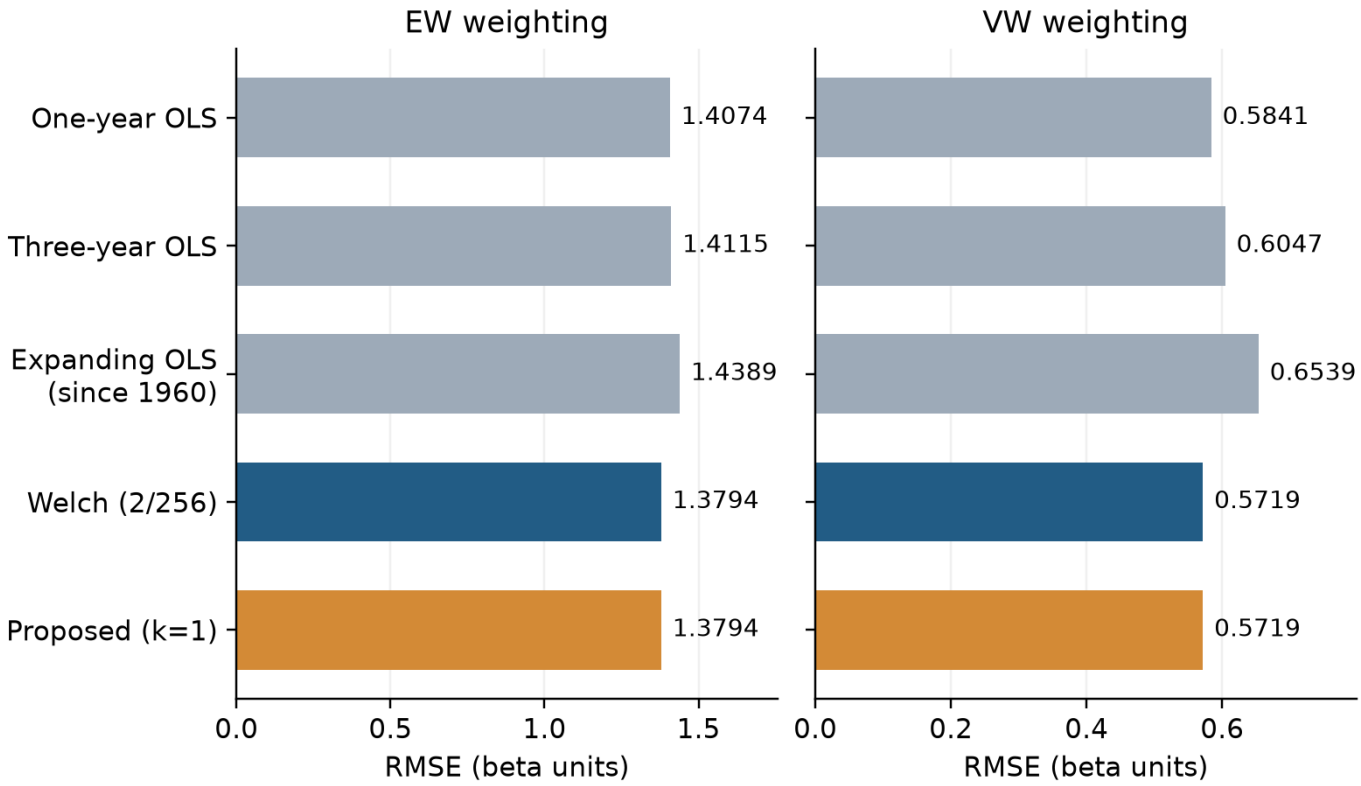
Statistic	ALL	IS	OOS
Weighted projection R2: mean (%)	31.691	32.777	30.115
Weighted projection R2: minimum (%)	15.562	19.703	15.562
Weighted projection R2: maximum (%)	53.801	50.248	53.801
Max relative orthogonality error	8.952e-15	8.952e-15	5.354e-15
Max absolute FWL slope error	4.267e-16	4.267e-16	2.628e-16
Max affine slope-mapping error	2.984e-15	2.984e-15	9.324e-16
Max affine fitted-return difference	2.225e-13	2.225e-13	4.702e-14

Notes: Dates match Table 6. The 750 monthly cross sections have identical stocks and sqrt(ME) objective weights in the signal projection, full return regression and pure-signal return regression. Controls are four non-value characteristics plus full represented industry indicators spanning the constant. The pure signal retains its projection-residual units. Weighted R2 describes cross-sectional signal variation explained, not return prediction. Relative orthogonality divides |C_j'W z_perp| by weighted column and signal norms. FWL error is full-signal slope minus pure-signal slope. Affine transformation is z_new=2*z+3 for all five characteristics; characteristic slopes halve. Fit differences are decimal monthly returns. Residual differences equal fitted-return differences in magnitude because y is unchanged; no separate saved residual-error series exists. Source: q3/FWL_diagnostics.csv.

Economic interpretation

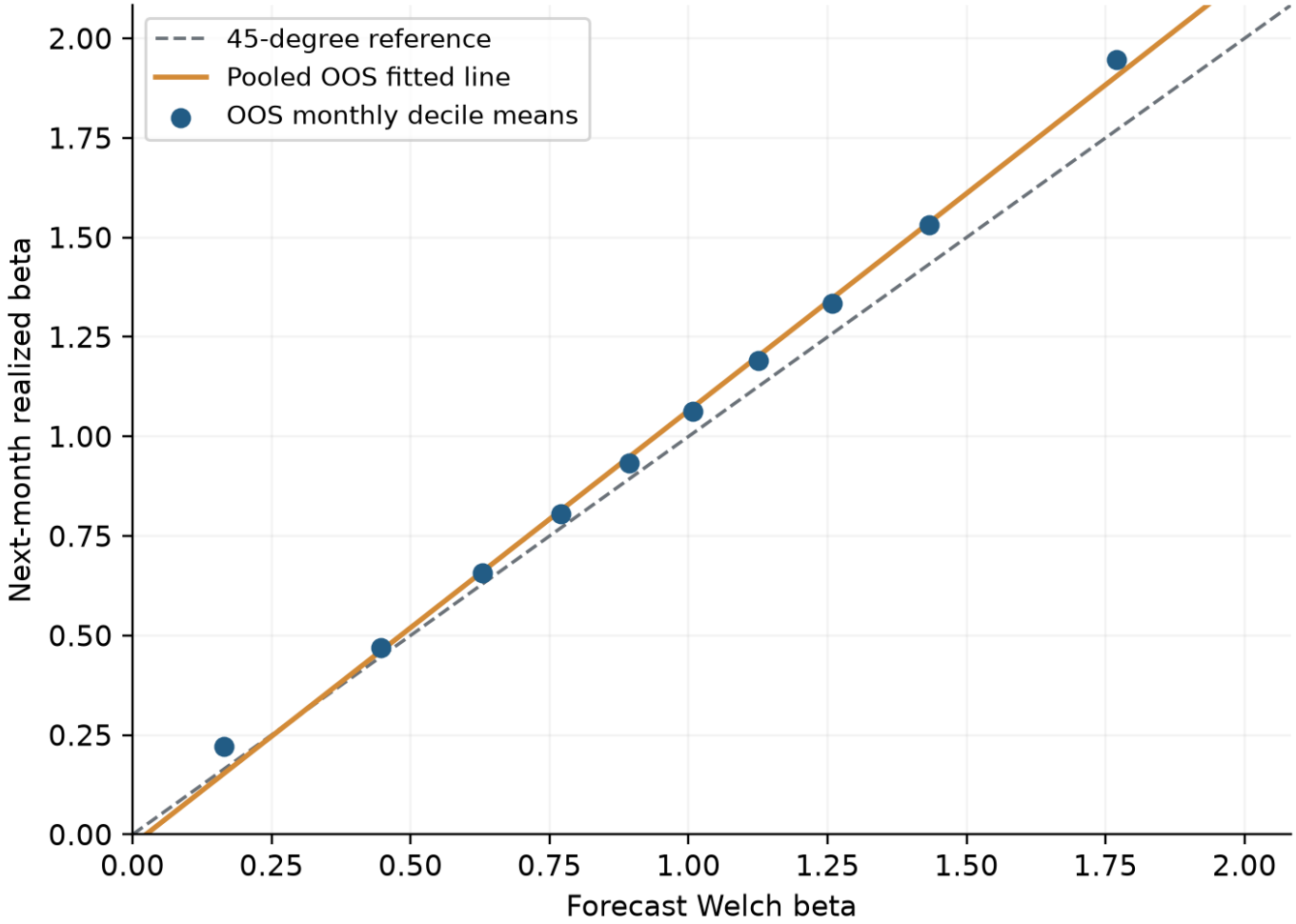
FWL slope equality and affine fitted-span invariance hold algebraically under the matched-row/weight and constant-span conditions. They do not establish a return premium. Omitting profitability changes the economic comparison rather than merely changing units. The full and pure return regressions need not share fitted returns or residuals.

Figure A. Q1 out-of-sample beta forecast accuracy



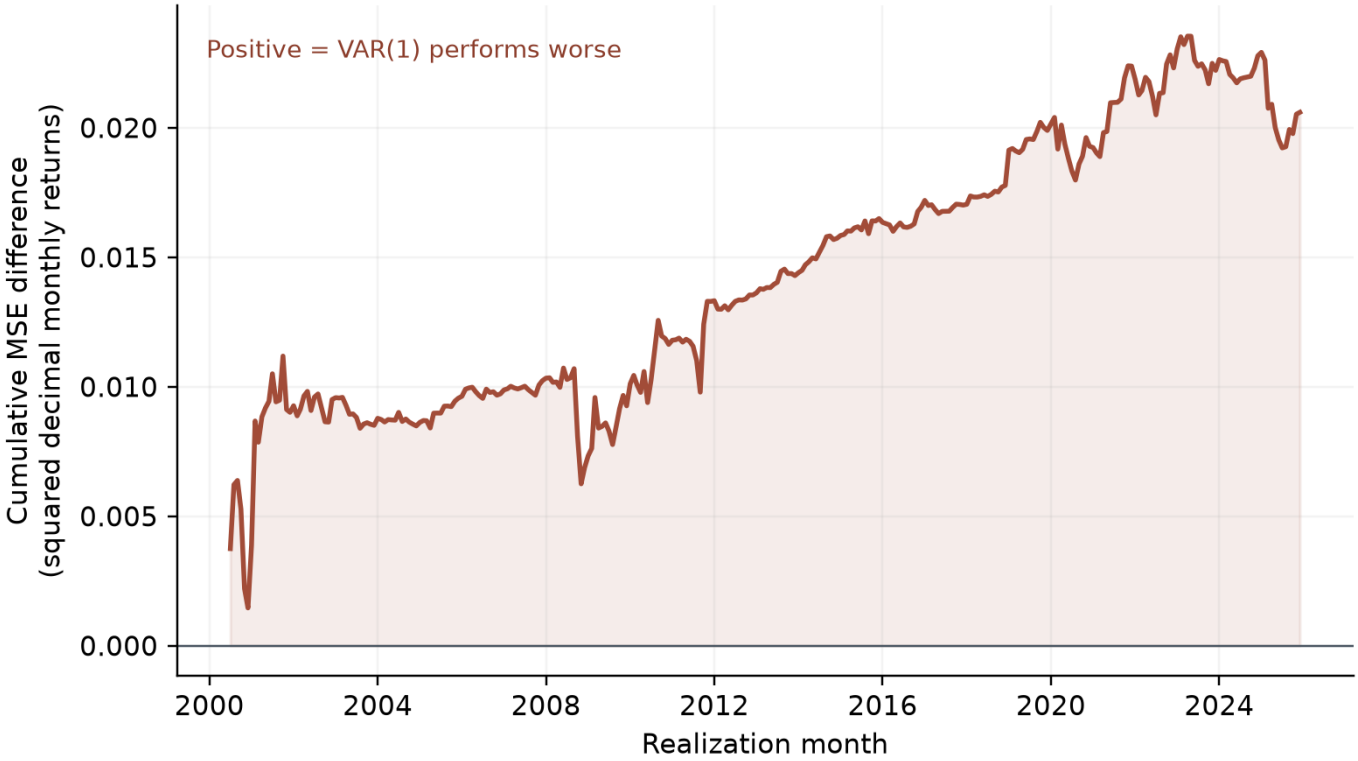
July 2000-December 2025; 306 months, 1,291,875 matched stock-months. Beta-unit RMSE averages normalized monthly MSE before taking the square root. Equal versus origin-ME weighting; identical stock-month keys across all estimators. Welch is rho=2/256, calendar aging. Expanding OLS starts in 1960. Proposed shrinkage k=1 equals Welch. Source: q1/rmse.csv.

Figure B. Welch calibration against realized beta



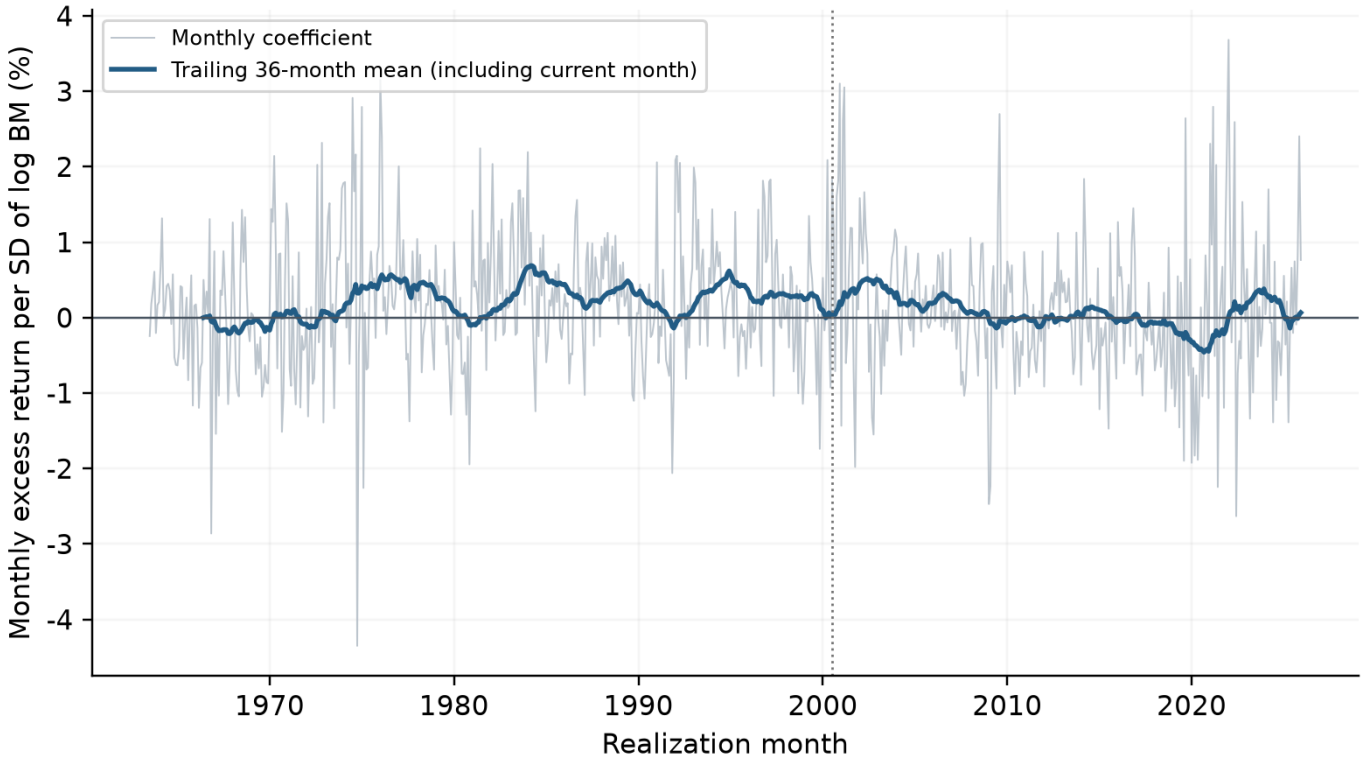
July 2000-December 2025; 306 months, 1,291,875 stock-months. Points average the saved monthly Welch-sorted decile means with equal month weights. The orange fitted line uses the saved pooled stock-month OLS intercept and slope, not a regression on these ten points. Gray dashes show the 45-degree reference. Welch rho=2/256 with calendar-session aging; betas are unitless. Sources: q1/beta_sorts.csv and pooled_calibration.csv.

Figure C. Cumulative out-of-sample stock forecast loss



July 2000-December 2025; 306 months, 1,155,444 matched stock-months. Each increment is saved monthly VW stock MSE for VAR(1) minus expanding factor means. Same three-factor betas, stocks, returns and origin-ME weights across forecasts. Cumulative sum begins at the first OOS monthly loss; it is not a portfolio return. Positive values indicate worse VAR performance. Units: squared decimal monthly returns. Source: q2/monthly_stock_losses_VW.csv.

Figure D. Monthly value coefficient and trailing average



July 1963-December 2025; 750 months, 2,668,791 matched stock-months. Thin gray line is the full-model monthly log-BM coefficient; blue is the trailing 36-month arithmetic mean, including the current month, requiring 36 observations. No centered or future window is used. Vertical dotted line marks July 2000. Coefficients are multiplied by 100: percent monthly excess return per cross-sectional SD of log BM. WLS uses sqrt(ME) objective weights and FF12 plus the other four characteristics. Source: q3/monthly_coefficients.csv.